

CLEARWATER BEACH PIER 60, FL, USA

WMO: 997345

Lat: 27.9777N

Lon: 82.8321W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.6	(c) 7.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.4	(k) 13.7	(l) 13.1	(m) 13.1	(n) 5.5	(o) 60	(p) 0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 3.9	(c) 31.3	(d) N/A	(e) 30.7	(f) N/A	(g) 30.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.4	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.0	(b) 11.2	(c) 9.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	15.5	17.0	19.1	21.9	24.8	27.2	27.8	28.4	27.2	24.4	19.8	17.9		
	DBStd	5.23	4.26	4.08	3.28	2.18	1.76	1.21	1.05	0.91	1.20	2.66	3.13	4.12		
	HDD10.0	14	9	3	0	0	0	0	0	0	0	0	0	3		
	HDD18.3	288	106	66	32	3	0	0	0	0	0	2	21	58		
	CDD10.0	4618	178	199	281	358	458	516	553	570	516	448	293	248		
	CDD18.3	1851	16	29	55	112	199	266	294	312	266	191	64	46		
	CDH23.3	16441	16	34	58	313	1372	2930	3458	3830	2786	1416	153	76		
(13)	CDH26.7	4624	0	2	2	14	146	817	1181	1460	782	213	7	1		
(14)	Wind	WSAvg	4.6	4.8	4.6	5.3	5.1	4.7	4.6	4.0	4.1	4.3	4.6	4.6		
(15) Precipitation	PrecAvg	1303	66	69	78	63	64	176	215	221	178	71	42	60		
	PrecMax	1812	138	285	223	219	212	354	380	350	348	174	140	391		
	PrecMin	1036	11	12	10	9	5	76	105	123	51	14	9	14		
	PrecStd	197	39	56	60	42	52	75	63	69	79	36	30	93		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	26.3	26.5	28.3	30.2	31.7	31.9	32.3	32.3	30.6	27.9	26.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	23.5	24.2	24.9	26.8	29.0	30.5	30.7	31.1	31.0	29.3	26.5	25.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	22.3	23.2	23.9	25.9	28.1	30.0	30.2	30.7	30.1	28.6	25.3	24.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	21.1	22.1	23.1	25.1	27.4	29.6	29.9	30.3	29.6	27.9	24.2	23.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	5.3	5.2	4.7	4.0	3.6	3.7	3.9	4.6	4.9	5.6	5.8		
		MCDBR	6.9	5.7	6.0	6.2	5.6	4.7	4.3	4.3	5.7	5.4	6.5	6.0		
	5% WB	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.358	0.369	0.393	0.424	0.449	0.473	0.455	0.423	0.390	0.355	0.345		
	taud		2.528	2.499	2.455	2.353	2.324	2.332	2.272	2.335	2.391	2.468	2.512	2.553		
	Ebn at Noon		900	915	922	904	871	844	822	837	857	869	880	884		
	Edh at Noon		90	100	111	126	130	128	136	127	116	102	91	85		
(44) All-Sky Solar Radiation	RadAvg		3.52	4.25	5.42	6.51	6.94	6.30	5.96	5.71	5.17	4.77	3.87	3.25		
	RadStd		0.26	0.30	0.48	0.33	0.37	0.50	0.35	0.35	0.37	0.30	0.22	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	-4	-48	+167	+115

Nomenclature: See separate page